

The majority of Earth's mass is composed of compounds belonging to the group known as ionic compounds. At the end of this chapter, you will be able to explain the structure and properties of these compounds.

Ionic compounds are made by the combination of positive and negative ions. You will see that the properties of ionic compounds are a direct result of the bonding between the particles within these compounds.

Writing chemical formulae and naming ionic compounds are other important skills that you will learn in this chapter.

Science understanding

- the ability of atoms to form chemical bonds can be explained by the arrangement of electrons in the atom and in particular by the stability of the valence electron shell
- the type of bonding within ionic, metallic and covalent substances explains their physical properties, including melting and boiling points, conductivity of both electricity and heat and hardness
- ionic bonding can be modelled as a regular arrangement of positively and negatively charged ions in a crystalline lattice with electrostatic forces of attraction between oppositely charged ions
- the ionic bonding model can be used to explain the properties of ionic compounds, including high melting point, brittleness and non-conductivity in the solid state; the ability of ionic compounds to conduct electricity when molten or in aqueous solution can be explained by the breaking of the bonds in the lattice to give mobile ions
- ions are atoms or groups of atoms that are electrically charged due to a loss or gain of electrons; ions are represented by formulae which include the number of constituent atoms and the charge of the ion (for example, O^{2-} , SO_4^{2-})
- the formulae of ionic compounds can be determined from the charges on the relevant ions

5.1 Properties and structures of ionic compounds



FIGURE 5.1.1 Many gemstones are made from ionic compounds.

In this chapter, you will study the structure and properties of a group of substances called **ionic compounds**. Ionic compounds are made by the combination of atoms, or groups of atoms, where electron(s) are transferred from one to another. In doing so, these particles become ions, hence ionic compounds are compounds formed from ions.

These materials are very common in the natural world because the Earth's crust is largely made up of complex ionic compounds. Most rocks, minerals and gemstones (Figure 5.1.1) are ionic compounds. Soil contains weathered rocks mixed with decomposed organic material, so soil contains large quantities of ionic compounds.

Ceramics, kitchen crockery and bricks are made from clays. Clays are formed by the weathering of rocks, so these materials also contain ionic compounds. Kitchen crockery and bricks contain mixtures of different ionic compounds. Table salt (sodium chloride) is a pure ionic compound.

PROPERTIES OF IONIC COMPOUNDS

If you think about the characteristics of rocks, kitchen crockery and table salt, you will recognise that these materials, and therefore ionic compounds, have some properties in common.

Table 5.1.1 lists some properties of typical ionic compounds. These compounds can be found in materials you might encounter in everyday life. Note that the compounds listed are simple ionic compounds, whereas rocks, ceramics and bricks contain more complex ionic compounds.

TABLE 5.1.1 Properties of typical ionic compounds

Ionic compound	Melting point (°C)	Conductive as solid	Conductive as liquid	Conductive in aqueous solution (0.1 mol L ⁻¹)	Solubility in water at 25°C (g/100 g water)	Example of commercially available product containing the compound
copper(II) sulfate	decomposes 110	no	yes	yes	22	bluestone spray (used to kill pathogens on fruit)
sodium chloride	801	no	yes	yes	36	food salt
calcium carbonate	1339*	no	yes	–	0.0013	main component in marble
zinc oxide	1975	no	yes	–	insoluble	zinc cream
sodium hydroxide	318	no	yes	Yes	114	oven cleaner

*Melting point determined under pressure to prevent decomposition of compound.

Lists of data about ionic compounds, such as those in the Table 5.1.1, have allowed chemists to summarise their properties.

Generally, ionic compounds:

- have high melting and boiling points and they are all solids at room temperature
- are hard but brittle (fragile), unlike metals and therefore they are neither malleable nor ductile
- do not conduct electricity in the solid state
- are good conductors of electricity in the liquid state or when dissolved in water
- vary from very soluble to insoluble in water.

DEDUCING THE STRUCTURE OF A COMPOUND FROM ITS PROPERTIES

You will remember from Chapter 4 that the properties of metals indicate something about the structure of metals; that is, how particles are arranged in a metallic lattice. The properties of ionic compounds also provide evidence for how the particles are arranged in ionic compounds.

From Table 5.1.1, you can see that some of the physical properties of ionic compounds are different from those of metals. Chemists have concluded that the particles in ionic compounds, and how they are arranged in the solid state, are different from those present in metals.

Table 5.1.2 lists the observed properties of one particular ionic compound, sodium chloride (table salt). Beside each property is a description of the nature of the particles and therefore the types of forces between the particles. These descriptions can be inferred from the different properties of sodium chloride.

TABLE 5.1.2 Properties of sodium chloride and the information this provides about its structure

Property	What this tells us about structure
high melting point	Forces between the particles are strong.
hard, brittle crystals	Forces between the particles are strong.
does not conduct electricity in the solid state	No free-moving charged particles are present in solid sodium chloride.
conducts electricity in the molten state	Free-moving charged particles are present in molten sodium chloride.

In summary, for ionic compounds generally:

- the forces between the particles are strong
- there are no free-moving electrons present, as there are in metals
- there are charged particles present, but in the solid state they are not free to move
- when an ionic compound melts, the charged particles are free to move and then the compound will conduct electricity.

THE IONIC BONDING MODEL

Now that you understand some of the details of the structure of ionic compounds, the next step is to work out how the particles in these compounds are arranged in the solid state.

When two atoms or groups of atoms react to form ionic compounds, the following steps occur.

- One set of atoms, generally metal atoms, lose electrons and so become positively charged ions (called cations).
- These lost electrons are transferred to another set of atoms.
- The other set of atoms, generally non-metal atoms, become negatively charged ions (called anions).

You will remember from Chapter 3, page xx, how electrons arrange themselves into shells around the nucleus. Atoms are at their most stable when they have a filled valence shell. Therefore, an atom's ability to form a cation or anion depends on how many electrons it needs to gain or lose to achieve this stable arrangement.

Cations and anions then arrange themselves in the following way:

- Large numbers of cations and anions combine to form a three-dimensional lattice (or **crystal lattice**).
- The three-dimensional lattice is held together strongly by electrostatic forces of attraction between the oppositely charged ions. The electrostatic force of attraction holding the ions together is called **ionic bonding**.

- In the case of sodium chloride, sodium (electron configuration 2,8,1) loses one electron, so it has a full valence shell (electron configuration 2,8). Losing one electron makes it positively charged and it is given the symbol Na^+ . Each chlorine atom normally has the electron configuration of 2,8,7. Therefore, it only needs to gain one electron to obtain a full valence shell (electron configuration 2,8,8). Gaining one electron makes it negatively charged and it becomes an anion, Cl^- .
- In sodium chloride, in order to maximise the forces of attraction, each sodium ion is surrounded by six chloride ions and each chloride ion is surrounded by six sodium ions, as shown in Figure 5.1.2.
- Even though each chloride ion is close to another chloride ion (Figure 5.1.2), the attractive force they have towards the sodium ions outweighs the repulsive force from the chloride ions so the lattice is held together quite strongly.

i Ionic bonding is the strong, electrostatic attraction that exists between positively charged cations and negatively charged anions.

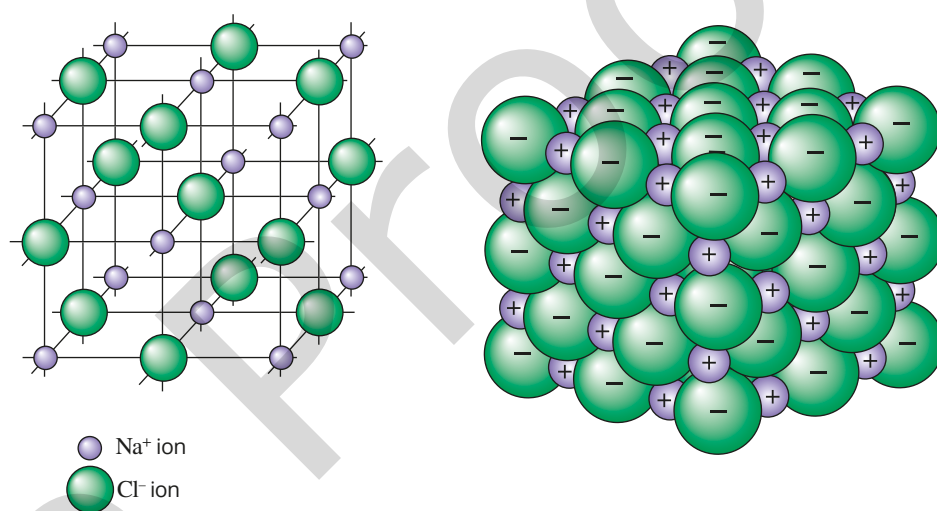


FIGURE 5.1.2 Two representations of part of the crystal lattice of the ionic compound sodium chloride. Forces of attraction between oppositely charged ions result in strong bonding.

The formula of sodium chloride

The **chemical formula** of sodium chloride is written as NaCl . However, it is important to note that in a solid sample of an ionic compound, such as sodium chloride, individual pairs of sodium and chloride ions do not exist. The solid is also not built up of discrete NaCl molecules.

Instead, the solid is made up of a continuous lattice of alternating Na^+ and Cl^- ions. NaCl is the formula unit for this compound; it represents the simplest whole number ratio of the ions that the substance is composed of. All sodium ions are an equal distance from six chloride ions and all chloride ions are an equal distance from six sodium ions. The overall ratio of sodium ions to chloride ions in the lattice is 1:1, so that is why the formula is written as NaCl .

5.1 Review

SUMMARY

- Ionic compounds form a crystal lattice made up of positive and negative ions.
- Cations are positive ions and are generally metals.
- Anions are negative ions and are generally non-metals.
- The three-dimensional lattice is held together strongly by electrostatic forces of attraction between the cations and anions. The electrostatic forces of attraction are called ionic bonding.
- In the case of sodium chloride, in order to maximise the forces of attraction, each sodium ion (Na^+) is surrounded by six chloride ions (Cl^-) and each chloride ion is surrounded by six sodium ions.
- Ionic compounds have high melting and boiling points. They are solids at room temperature.
- Ionic compounds are hard but brittle. Unlike metals, they are neither malleable nor ductile.
- Ionic compounds do not conduct electricity in the solid state.
- Ionic compounds are good conductors of electricity in the liquid state or when dissolved in water.
- In water, ionic compounds vary from very soluble to insoluble.

KEY QUESTIONS

- 1 Some properties of four different substances are described below. Which substance is most likely to be an ionic compound?
 - A Substance A has a melting point of 842°C and conducts electricity at 700°C .
 - B Substance B has a melting point of 308°C . It does not conduct electricity at 250°C but will conduct electricity at 350°C .
 - C Substance C has a melting point of 180°C and can be drawn into a wire.
 - D Substance D is a white solid that melts at 660°C and will not conduct electricity at 700°C .
- 2 Sodium chloride does not conduct electricity in the solid state but does conduct when molten (liquid).
 - a How could you use Figure 5.1.2 on page xx to explain why solid sodium chloride does not conduct electricity?
 - b Explain why molten sodium chloride conducts electricity.
- 3 Explain why aluminium would be more likely to form a cation than an anion.
- 4 Explain why the crystal lattice is held together so strongly in sodium chloride, even though the number of cations is equal to the number of anions.

5.2 Using the ionic bonding model to explain properties



FIGURE 5.2.1 Bricks made from the ionic compound magnesium oxide are used to line furnaces and kilns.



FIGURE 5.2.2 Ceramic brake discs work more effectively than steel ones at high temperatures. A ceramic brake disc contains ionic compounds that have very high melting temperatures and withstand the heat produced by braking better than metals.

In Chapter 4, you saw that the metallic bonding model represents the structure of metals as a lattice of positively charged metal ions held together by delocalised electrons. This model explains many of the properties of metals, such as why metals generally have high melting points and conduct electricity in the solid state.

In this section, you will see how the ionic bonding model explains the properties of ionic compounds.

HIGH MELTING POINTS

To melt an ionic solid such as sodium chloride, you must provide energy to allow the ions to break free and move. Sodium chloride has a high melting point (801°C). This indicates that a large amount of energy is needed to overcome the electrostatic attraction between oppositely charged ions and allow them to move freely. Therefore, the ionic bonds between the positive sodium ions and negative chloride ions must be strong, which explains why a high temperature is required to melt solid sodium chloride.

The high melting point of ionic compounds is put to use in the bricks that line furnaces and kilns (Figure 5.2.1) and in the ceramic materials used to make brake discs for high-performance cars (Figure 5.2.2).

HARDNESS AND BRITTLINESS

There are strong electrostatic forces of attraction between ions in an ionic compound, so a strong force is needed to disrupt the crystal lattice. Therefore, one of the properties of ionic compounds is that they are hard. This means that a sodium chloride crystal cannot be scratched easily.

The strength of house bricks, concrete bridges and cobbled streets can be attributed to the ionic bonding within their structures.

Although a salt crystal is hard, a strong impact such as a hammer blow will shatter the crystal. Therefore, it is said to be brittle. This is because the layers of ions will move relative to each other due to the force of the blow.

CHEMFILE

Early tools

Some of the earliest tools used by humans were axes, spearheads and coarse needles used for weaving. Each application required a material that was hard and could be shaped. Certain types of rocks that are composed of ionic compounds served this purpose well.

Figure 5.2.3 shows a primitive axe used during the Stone Age. The Stone Age ended at different times in different parts of the world, as humans learnt to smelt (fuse or melt) metals such as copper from their ores to create more refined and lighter tools. Their metallic nature meant that they were not as hard as the original tools (made of rock composed of ionic compounds).



FIGURE 5.2.3 The hardness of ionic compounds is the reason why axes were once made from rocks.

During this movement, ions of like charge are shifted so they are next to each other, as seen in Figure 5.2.4. The resulting repulsion between the similarly charged ions causes the crystal to shatter.

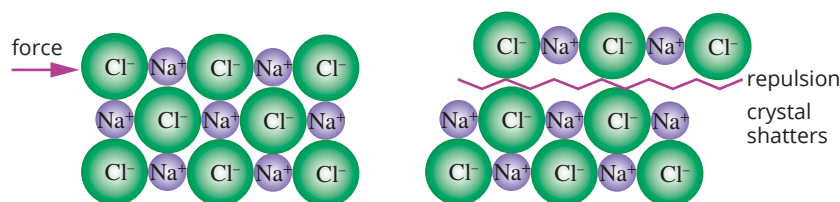


FIGURE 5.2.4 A lattice of an ionic compound shattering. Note that just before shattering, the Cl^- ions are adjacent to other Cl^- ions and the Na^+ ions are also next to each other.

Materials made from clay, such as kitchen crockery (Figure 5.2.5), ceramic tiles and bricks are hard, but they are also brittle.

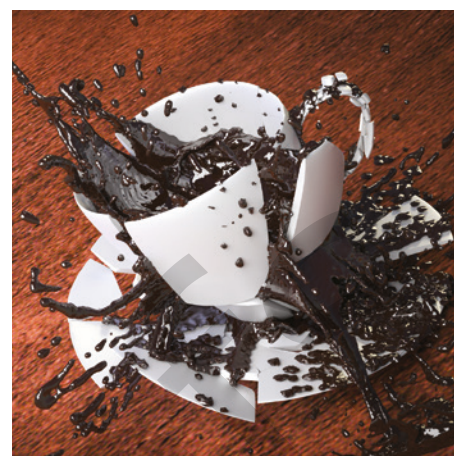


FIGURE 5.2.5 A coffee cup being smashed

CHEMFILE

Porcelain

Porcelain is a type of chinaware made from a clay called kaolin. Figure 5.2.6 shows a porcelain dinner setting. Clay is weathered rock and consists of a mixture of complex ionic compounds. The chemical formula for kaolin is $\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4$. The ions in kaolin are aluminium (Al^{3+}), silicate ($\text{Si}_2\text{O}_5^{2-}$) and hydroxide (OH^-).

Porcelain is made by moulding the object from kaolin and then heating it to about 1300°C in a kiln. Because it is made from ionic compounds, the resulting porcelain

is hard but brittle. Be careful not to drop the family heirloom chinaware as it will most likely shatter!



FIGURE 5.2.6 Porcelain cups and plates are made from clay, which contains ionic compounds. Therefore, porcelain is hard and brittle.

ELECTRICAL CONDUCTIVITY

In the solid form, ions in sodium chloride are held in the crystal lattice and are not free to move, so solid sodium chloride does not conduct electricity. Remember that for a substance to conduct electricity, it must contain charged particles that are free to move. Figure 5.2.7 shows how the particles are arranged in an ionic compound in solid form.

The force of attraction between oppositely charged ions is strong, so ionic compounds are hard and have high melting points.

In the solid state oppositely charged ions are held strongly within the lattice and cannot move. Solid ionic compounds do not conduct electricity.

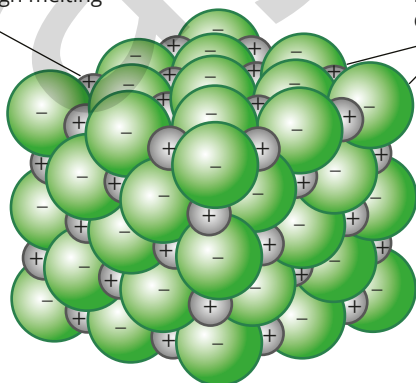


FIGURE 5.2.7 The arrangement of ions within a solid ionic compound form a crystal lattice.



FIGURE 5.2.8 A ceramic insulator on the post of an electric fence

The non-conducting property of ionic compounds is used in ceramic insulators, which are used to keep high-voltage power lines insulated from electricity poles and electric fence wires (Figure 5.2.8).

When solid ionic compounds melt, the ions become free to move, enabling the cations and anions in the molten compound to conduct electricity.

Similarly, when ionic compounds dissolve in water, ionic bonds in the lattice are broken and the ions are separated and move freely in solution.

When an electric current is applied to either a molten ionic compound or a solution of the compound in water, positive ions move towards the negatively charged electrode (cathode) and negative ions move towards the positively charged electrode (anode), resulting in an electric current as shown in Figure 5.2.9.

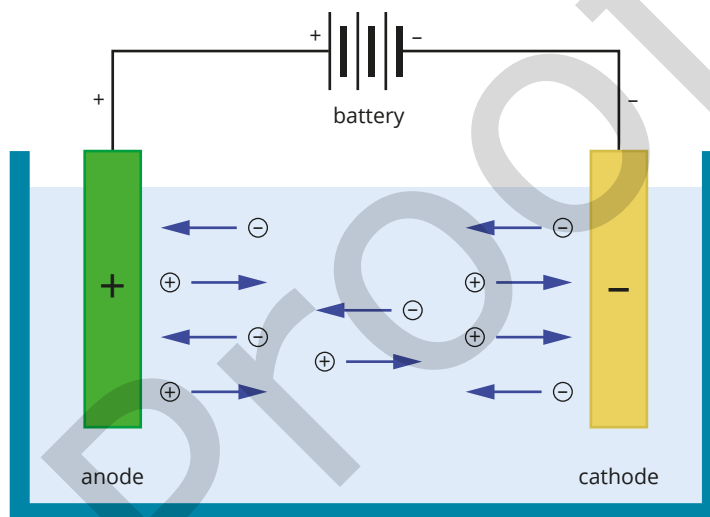


FIGURE 5.2.9 A molten ionic compound will conduct an electric current.

EXTENSION

Electrolysis

When an electric current passes through a molten ionic compound, or a solution of the compound, chemical reactions occur at the positive and negative electrodes. This process is known as electrolysis and is used to make a variety of chemicals. Chemicals formed by electrolysis are often difficult to obtain by other means.

For example, the final stage in the industrial production of aluminium involves passing an electric current through a molten liquid containing aluminium oxide (alumina). Aluminium metal is deposited at the negatively charged electrode.

Sodium metal is a starting material for the manufacture of a range of organic chemicals, including dyes. It is also produced on a large scale by electrolysis. In this case, electricity is passed through molten sodium chloride to make sodium. During this process, sodium ions move towards the negative electrode where they become sodium metal, and chloride ions move to the positive electrode and are converted into chlorine gas (Figure 5.2.10).

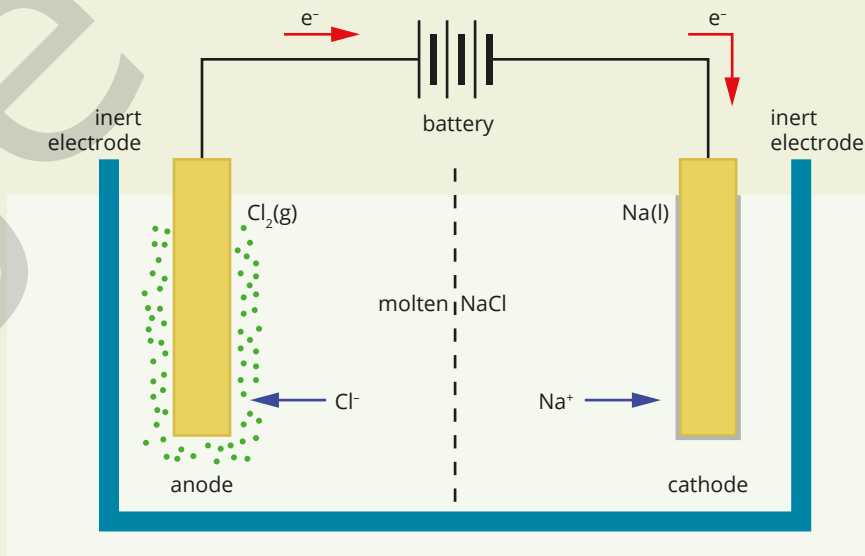


FIGURE 5.2.10 The electrolysis of molten sodium chloride produces sodium metal and chlorine gas.

i For an object to conduct electricity, it must contain charged particles that are free to move. Solid ionic compounds are made up of a crystal lattice, so the particles are not free to move and they cannot conduct electricity. When molten (melted) or dissolved in water, the charged particles are free to move and conduct.

A solution or molten substance that conducts electricity by means of the movement of ions is called an **electrolyte**.

SOLUBILITY

Some ionic compounds are very soluble in water, whereas others are very insoluble. When a soluble ionic compound is added to water, the ions break away from the ionic lattice and mix with the water molecules. If an insoluble compound is added to water, the ions remain bonded together in the ionic lattice and do not form a solution.

Whether an ionic compound is soluble or insoluble depends on the relative strength of the forces of attraction between the:

- positive and negative ions in the lattice
- water molecules and the ions.

You will look at the solubility of ionic compounds in water in more detail in Chapter 15.

USES OF IONIC COMPOUNDS

Ionic compounds have a wide variety of uses. Some of these uses are related directly to the physical properties of the compounds, such as hardness, high melting points and their ability to conduct electricity in solution. A number of ionic compounds and their uses in relation to their characteristic properties are described below.

Hardness

- Calcium phosphate is a constituent of bone tissue that gives it strength.
- Calcium sulfate in the form of gypsum is used to make plasterboard for lining the walls and ceilings of houses.
- Granite, limestone and sandstone are used as building stone.
- Bricks, tiles and crockery are made from clay, a material that contains particles held together by strong ionic bonding.

High melting point

- Magnesium oxide and other ionic compounds are used to line furnaces.
- Ceramics are materials that contain a mixture of strong ionic and covalent bonds. They are used in some engine parts.

Electrical conductivity

- Ammonium chloride is used as an electrolyte in dry cell batteries. Electrolytes contain ions in solution and this allows a current to flow in the battery.
- Potassium hydroxide is used as an electrolyte in the 'button' cells used in small electronic devices such as watches and calculators.

Other uses of ionic compounds

- Sodium hydrogencarbonate (bicarbonate of soda or baking soda) is used in baking to cause cakes to rise when placed in a hot oven. This occurs because the compound decomposes when heated to produce carbon dioxide gas.
- Sodium chloride has many uses, including as a flavouring agent and preservative in food.
- Sodium hypochlorite is used as a bleach and in swimming pools to kill microorganisms.

CHEMISTRY IN ACTION

Salt of the Earth

Salt is added to many foods as a preservative or to 'improve' the flavour. It is rarely essential to add salt—a diet that has meat and milk provides enough sodium for our needs (Tables 5.2.1 and 5.2.2). It is the sodium part of sodium chloride that we should limit in our diets.

TABLE 5.2.1 The sodium content of some natural foods

Food	Sodium content (g Na/100 g food)
beef	0.06
chicken	0.1
fish	0.1
crab meat	0.35
rice	–
cow's milk	0.06
celery	0.15
silverbeet	0.7
fresh fruit	–
most vegetables	0.1

TABLE 5.2.2 The sodium content of some processed foods

Food	Sodium content (g Na/100 g food)
sausage	1.2
ham	1.1
cheddar cheese	0.7
meat pie	0.5
tomato sauce	1.2
soy sauce	7.0
bread	0.5
potato chips	1.0
olives	2.3
salted nuts	4.0
doughnuts	0.5
ice-cream	0.08

A healthy adult requires about 2 grams of sodium per day to replace the sodium they excrete in urine and lose when they perspire (sweat). Salt plays a role in the passage of nerve impulses and helps control the movement of water in and out of cells. A salt deficiency can lead to muscular cramps, loss of appetite and reduced brain function.

Doctors are increasingly critical of the amount of salt in our diets because of the connection between dietary salt and hypertension, or high blood pressure. The average Australian consumes 11 grams of salt daily. When you consider that a take-away meal can contain about 3.5 grams of salt, it is easy to see how you could reach this level. Food manufacturers have responded to public demand by decreasing the salt content in processed foods.

The body attempts to maintain the salt concentration within narrow limits. After a salty meal, you feel thirsty—your body's response to the increased salt concentration in the blood. You drink water to dilute the salt. This excess fluid remains in your body until the kidneys have removed the excess salt.

Ancient civilisations recognised that we need salt in our diets. Settlements were based in regions where salt occurred naturally or on salt trade routes. Wars were even fought over salt, with Roman soldiers being paid a salt allowance (the word 'salary' comes from the Latin word for salt). Salt is harvested from salt lakes, salt mines or the sea (Figure 5.2.11). Solar energy is commonly used to extract salt from seawater.



FIGURE 5.2.11 Harvesting salt from evaporation lakes

Salinity

Salinity is the presence of salt in groundwater and soil. Plants, like animals, can only tolerate a limited range of salt concentrations. Some plants growing on sand dunes tolerate higher amounts of salt than others. However, if the salt concentration in the soil is too high, even those plants die.

Salinity is caused by increasing amounts of groundwater (water found under the surface of the Earth). As groundwater increases, the water table (water level) rises, bringing dissolved salts, in this case sodium chloride, closer to the surface. The increased salinity of water near the soil surface leads to poor plant growth and reduces the productivity of agricultural land that is used for grazing animals and growing crops.

The rise in the water table has been caused by excess water from irrigation flowing down through the soil, and the clearing of trees, which would otherwise have absorbed this water through their roots.

Reducing salinity

A number of organisations are working to reduce soil and water salinity. Strategies include:

- using irrigation water more efficiently
- improving drainage on the surface of the land
- improving drainage under the surface

- growing trees to soak up excess water
- sealing irrigation channels to prevent leakage.

These are long-term projects, although reduced salinity is apparent in some areas already.

One property of salt that has the potential to cause environmental damage is its ability to dissolve in water (Figure 5.2.12).



FIGURE 5.2.12 Land that is affected by salinity

5.2 Review

SUMMARY

- Ionic compounds are hard and have high melting and boiling points. This is because of the strong forces of attraction between the positively and negatively charged ions in the ionic lattice.
- When an ionic compound is hit, the ions move within the lattice so that like-charged ions line up opposite each other and then repel, causing the lattice to be disrupted. This makes ionic compounds brittle.
- Ionic compounds do not conduct electricity in the solid state. Although the solid ionic lattice contains charged particles, the particles are not free to move.
- When ionic compounds are dissolved in water or are in molten form, the charged particles are free to move, which means they can conduct electricity.
- In water, ionic compounds vary from very soluble to insoluble. The solubility depends on whether the forces between the water molecules and the ions in the lattice are strong enough to pull the ions out of the lattice.
- Ionic compounds are useful because of their physical properties such as hardness and high melting points.

KEY QUESTIONS

- 1 Why do ionic compounds have such high melting and boiling points?
- 2
 - a Explain why the lattice breaks apart in an ionic compound when hit with a heavy blow.
 - b How does this give ionic compounds their brittle property?
- 3 Use the ionic bonding model to explain why ionic compounds that have been heated above their melting point conduct electricity but will not when in solid form.
- 4 Sodium chloride or common salt is an essential part of our diet. Throughout history, it has also been used to preserve food in the absence of refrigeration. So important was sodium chloride to daily life that many words or expressions in the English language have been derived from the word salt. 'Salary' and 'salinity' are two of them. What others can you think of?

5.3 Formation of ionic compounds

Some of the reactions that occur between metals and non-metals to form ionic compounds are very vigorous. The reaction between sodium and chlorine to form sodium chloride produces a lot of heat. You will remember from Chapter 4 that sodium is very reactive. At high temperatures, the production of sodium chloride from sodium metal and chlorine gas is very explosive, producing a flame and large amounts of energy.

In the previous sections, you saw that the positive and negative ions that are formed in this reaction are arranged to form a three-dimensional lattice. In this section, you will learn how these positive and negative ions are formed from the atoms. You will also learn how the ratio of each type of atom in the compound is determined.

FORMING IONS

When forming an ionic compound, two things occur.

- One group of atoms loses electrons to form positively charged ions (cations). These are generally metals.
- Another group of atoms gains electrons to form negatively charged ions (anions). These are generally non-metals.

From Chapter 3, you will remember that most metals have low ionisation energies and low electronegativities. This means that the valence electrons on a metal atom are weakly held and can be removed easily.

Non-metals, on the other hand, have a higher ionisation energy and higher electronegativity. This means that non-metal atoms have a high affinity for valence electrons and they are not easily removed.

This is why in the reaction that forms many ionic compounds, it is the non-metal atom that takes one or more valence electrons from the metal atoms.

Most importantly, the ions that are formed usually have a full valence shell—a stable electron configuration. Most of the main group elements can accommodate eight electrons in their valence shell. So in the process of forming ions, both positive and negative, the tendency is for ions to end up with eight electrons in their outer shell (valence shell). You will recall that this is known as the octet rule.

Noble gases (group 18) are elements that already have the most stable valence shell configuration. Therefore, another way of thinking about the octet rule is that atoms tend to gain or lose electrons to gain a stable electron configuration identical to that of the noble gas nearest to them on the periodic table. The formation of stable ions is a powerful driving force in reactions between metals and non-metals when they produce ionic compounds.

For example, when sodium reacts with chlorine, each sodium atom loses one electron and each chlorine atom gains one electron.

After the reaction, the:

- sodium ion has the stable electron shell configuration of 2,8 (the same as a neon atom)
- chloride ion has the stable electron shell configuration of 2,8,8 (the same as an argon atom).

Figure 5.3.1 illustrates how, when sodium reacts with chlorine, an electron is lost by a sodium atom and gained by a chlorine atom. A diagram of this type is called an **electron transfer diagram**.

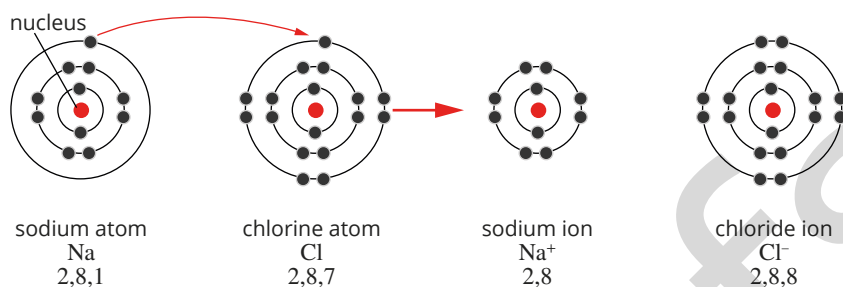


FIGURE 5.3.1 Electron transfer diagram showing the formation of sodium and chloride ions

The reaction between lithium and oxygen atoms is illustrated in Figure 5.3.2. In this reaction, an oxygen atom needs to gain two electrons to have eight electrons in its outer shell and form a stable ion. To allow this to happen, one oxygen atom will react with two lithium atoms, taking one electron from each atom.

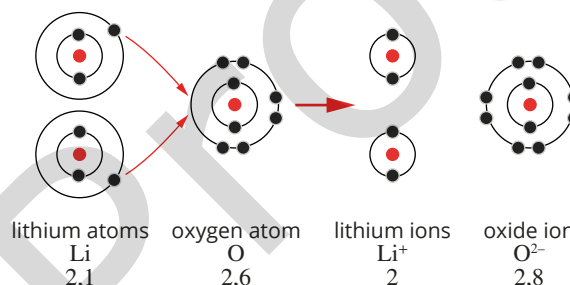


FIGURE 5.3.2 An electron transfer diagram showing the formation of lithium and oxide ions

After the reaction, there are just two electrons in what is now the outer shell of the Li⁺ ion, which is the same as the electron configuration of a helium atom.

In Chapter 3, you learned that atoms of elements in group 1 of the periodic table have electron configurations with one electron in the valence shell. Therefore, elements in this group often form cations with a charge of +1 as they readily lose this one outer-shell electron. For example, potassium has an electron configuration of 2,8,8,1. It will readily lose the one electron in the valence shell to have an electron configuration similar to that of argon (the nearest noble gas). The potassium ion now has an electron configuration of 2,8,8.

Metals in group 2 of the periodic table have electron configurations with two electrons in their valence shells. Therefore, they readily form ions with a charge of +2 as they lose these electrons. Magnesium is a metal in group 2 of the periodic table. The loss of the two valence electrons from an atom of magnesium is shown in Figure 5.3.3.

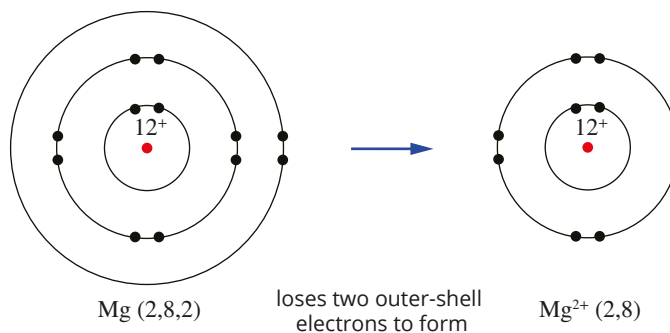


FIGURE 5.3.3 When two valence electrons are lost from a magnesium atom, the Mg²⁺ ion formed has a complete outer electron shell that contains eight electrons.

EXTENSION

Beryllium is an exception in group 2

Beryllium is found in group 2 of the periodic table. However, beryllium behaves very differently from the other alkaline earth metals in this group.

Beryllium has an electron configuration of 2,2. It could lose two electrons to form an ion similar to lithium with just one remaining valence shell (that is, full with two electrons). However, it does not do this because beryllium has a very high electronegativity compared with the rest of the group 2 metals. It tends to hold on to these two electrons. Beryllium mostly forms covalent compounds (which you will learn about in Chapter 6). For example, beryllium covalently bonds with two chlorine atoms to form beryllium chloride (BeCl_2). Beryllium chloride is used in the production of beryllium by electrolysis.

Non-metals in group 16 have six electrons in their valence shells, therefore they gain two electrons to form anions with a charge of -2 .

Group 17 non-metals have seven electrons in their valence shell. They readily gain one electron to fill the valence shell according to the octet rule. This means they form anions with a charge of -1 .

i Atoms are at their most stable when they have a full valence shell. Most of the main group elements can accommodate eight valence electrons. This is known as the octet rule.

ELECTRON TRANSFER DIAGRAMS

Electron transfer diagrams can be used to show how electrons are transferred from metallic atoms to non-metallic atoms to form an ionic compound. Figure 5.3.4 shows such a diagram for the reaction of magnesium with oxygen.

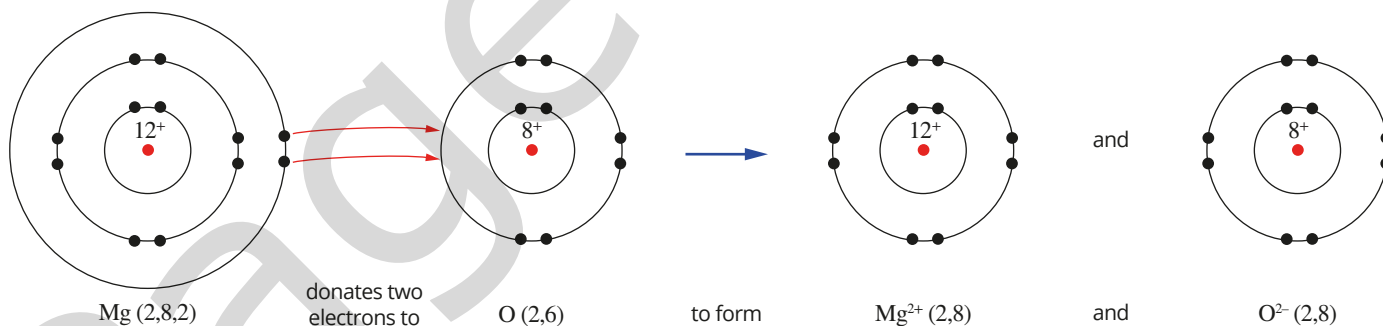


FIGURE 5.3.4 An electron transfer diagram showing the transfer of electrons when magnesium atoms react with oxygen atoms

i When an atom loses electrons, it becomes more positively charged as the number of protons no longer equals the number of electrons. The ion formed is written with a superscript $+$ sign indicating the charge. If there are two electrons lost it is written as $2+$. When an atom gains electrons, in the case of anions, they become negatively charged. This is written with a superscript $-$. For example, if three electrons are gained, the charge is written as $3-$ in superscript.

Figure 5.3.5 shows the diagram for the reaction of magnesium with chlorine. A magnesium atom has an electron configuration of 2,8,2 so it will readily lose its two outer-shell electrons. A magnesium ion (Mg^{2+}) is formed so that it now has the electron configuration of the nearest noble gas, neon (2,8). Each chlorine atom has an electron configuration of 2,8,7, so each will gain one outer-shell electron. A chloride ion (Cl^-) has the electron configuration of the noble gas argon (2,8,8). Because magnesium readily donates two electrons, it is found with two chlorine atoms, each chlorine atom taking one electron each from magnesium.

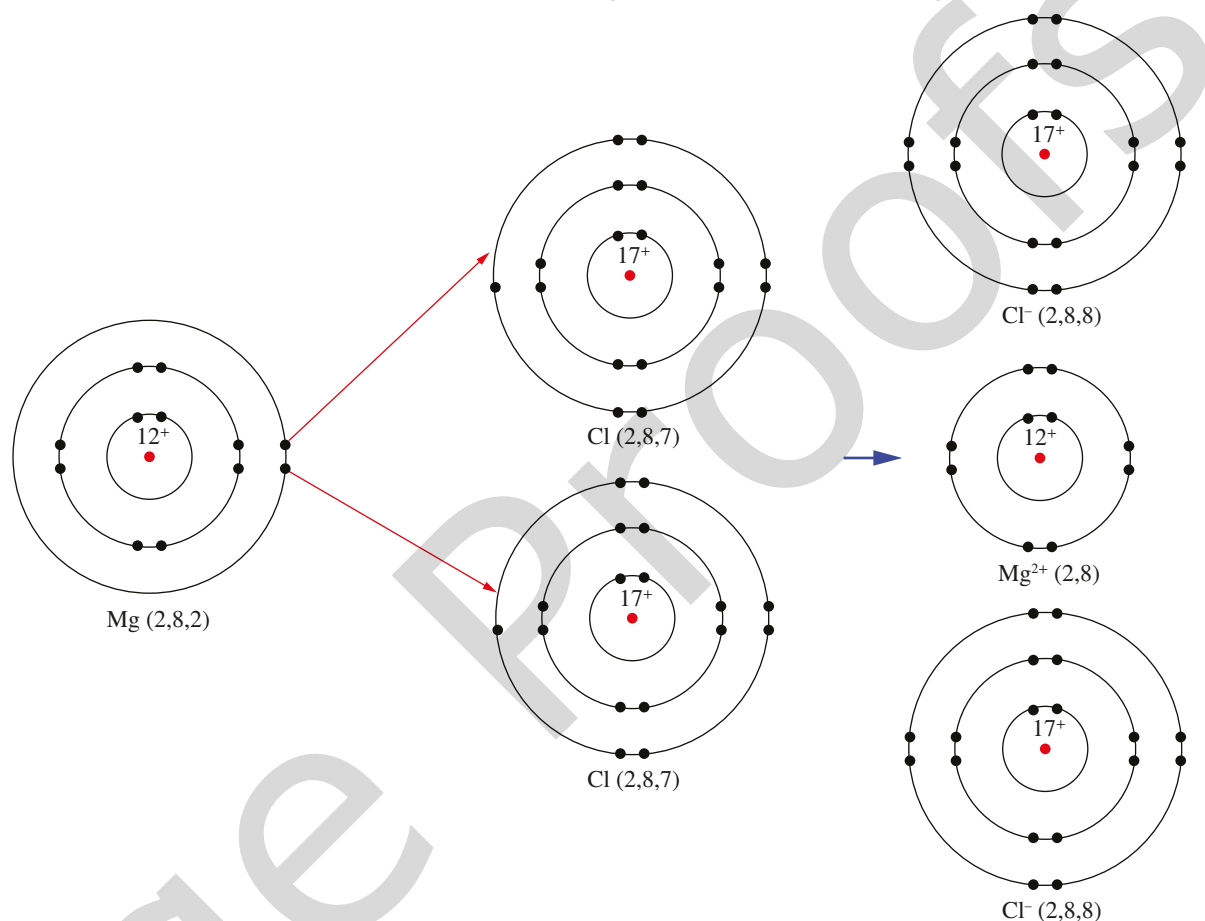


FIGURE 5.3.5 Electron transfer diagram showing the formation of the ionic compound magnesium chloride

To make sure you understand how ions are formed when metals and non-metals react, it can be useful to write equations for reactions that include the electron configurations of the reactants and products, as shown in Worked example 5.3.1.

Worked example 5.3.1

WRITING EQUATIONS FOR REACTIONS BETWEEN METALS AND NON-METALS ATOMS

Write an equation for the reaction between lithium and nitrogen atoms. Show the electron configurations for each element before and after the reaction.	
Thinking	Working
Write the symbol and electron configuration for the metal atom.	Li (2,1)
How many electrons will the metal atom lose from its outer shell when it reacts?	1
Write the symbol and electron configuration of the metal ion that will be formed.	Li ⁺ (2)
Write the symbol and electron configuration for the non-metal atom.	N (2,5)
How many electrons will the non-metal atom gain in its outer shell when it reacts?	3
Write the symbol and electron configuration of the non-metal ion that will be formed.	N ³⁻ (2,8)
The total number of electrons lost by metal atoms must equal the total number of electrons gained by non-metal atoms. What is the lowest number ratio of metal atoms to non-metal atoms that will allow this to happen?	metal atom : non-metal atom = 3:1
Using the ratio of metal ion : non-metal ion calculated above, write a balanced equation for the reaction. Show the electron configurations for both the reactant atoms and the product ions.	$3\text{Li (2,1)} + \text{N (2,5)} \rightarrow 3\text{Li}^+ (2) + \text{N}^{3-} (2,8)$

Worked example: Try yourself 5.3.1

WRITING EQUATIONS FOR REACTIONS BETWEEN METALS AND NON-METALS ATOMS

Write an equation for the reaction between calcium and phosphorus atoms. Show the electron configurations for each element before and after the reaction.

5.3 Review

SUMMARY

- During the formation of ionic compounds, metal atoms lose electrons to form positively charged ions (cations).
- During the formation of ionic compounds, non-metal atoms gain electrons to form negatively charged ions (anions).
- The ions present in an ionic compound have a stable electron configuration identical to that of the noble gas nearest to them on the periodic table.
- Electron transfer diagrams can be used to represent the formation of an ionic compound from its elements.
- When an ionic compound is formed from positively charged metal ions and negatively charged non-metal ions, the ions combine in proportions that produce an ionic compound with an overall zero charge.

KEY QUESTIONS

- 1 Indicate whether the following atoms will form cations or anions and explain why.
 - a calcium
 - b nitrogen
 - c fluorine
 - d aluminium
 - e phosphorus
- 2 Use diagrams similar to Figure 5.3.1 on page xx to show the formation of ions in the reactions between:
 - a potassium and fluorine
 - b magnesium and sulfur
 - c aluminium and fluorine
 - d sodium and oxygen
 - e aluminium and oxygen.
- 3 Why are group 2 metals of the periodic table likely to form cations with a charge of +2?
- 4 Explain why potassium chloride has the formula KCl, whereas the formula of calcium chloride is CaCl_2 .
- 5 Using the technique demonstrated in Worked example 5.3.1 on page XXX, write an equation for the reaction between the following metal and non-metal atoms. Show the electron configurations for each element before and after the reaction.
 - a sodium and chlorine atoms
 - b magnesium and oxygen atoms
 - c aluminium and sulfur atoms

5.4 Chemical formulae of simple ionic compounds

You have seen that ionic compounds contain oppositely charged ions that are arranged in three-dimensional lattices. The ions can have different charges. For example, aluminium forms ions with a +3 charge, whereas oxygen forms ions with a -2 charge.

In this section, you will learn how to use your knowledge of the charges on ions to write an overall formula for an ionic compound.

WRITING THE FORMULA OF AN IONIC COMPOUND

Because ionic compounds are electrically neutral, the total number of positive charges on the metal ions must equal the total number of negative charges on the non-metal ions. This is the most important guiding principle when you are trying to work out the formula of an ionic compound.

This can be seen with the formula of the ionic compound sodium chloride.

- A sodium ion (Na^+) has a +1 charge.
- A chloride ion (Cl^-) has a -1 charge.
- Therefore, in a crystal of sodium chloride, the ratio of sodium ions to chloride ions is 1:1 and the formula of sodium chloride is NaCl .

Using the same steps, you can work out the formula of magnesium chloride.

- A magnesium ion (Mg^{2+}) has a +2 charge.
- A chloride ion (Cl^-) has a -1 charge.
- Therefore, in a crystal of magnesium chloride, two chloride ions are needed to provide two negative charges so that they balance the +2 charge on every magnesium ion. Therefore, the ratio of magnesium ions to chloride ions in the crystal is 1:2 and the formula of magnesium chloride is MgCl_2 .

Figure 5.4.1 illustrates how formulae for some other ionic compounds can be determined.

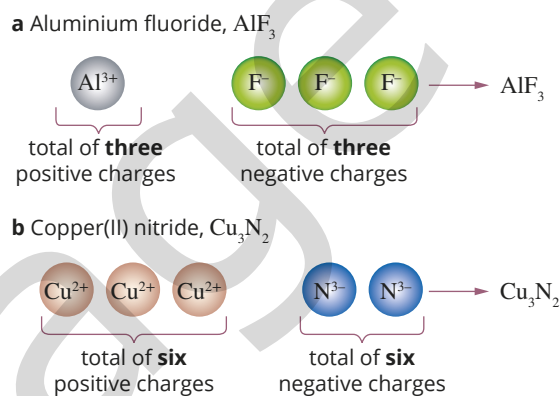


FIGURE 5.4.1 How to deduce chemical formulae from the charges on ions

Tables 5.4.1 and 5.4.2 list some of the more common positively and negatively charged ions. You may use these when you are writing formulae for ionic compounds.

TABLE 5.4.1 Names and formulae of some common cations

Charge		
+1	+2	+3
caesium, Cs ⁺ hydrogen, H ⁺ lithium, Li ⁺ potassium, K ⁺ rubidium, Rb ⁺ silver, Ag ⁺ sodium, Na ⁺	barium, Ba ²⁺ calcium, Ca ²⁺ cobalt(II), Co ²⁺ copper(II), Cu ²⁺ iron(II), Fe ²⁺ lead(II), Pb ²⁺ magnesium, Mg ²⁺ manganese(II), Mn ²⁺ nickel, Ni ²⁺ strontium, Sr ²⁺ zinc, Zn ²⁺	aluminium, Al ³⁺ chromium(III), Cr ³⁺ iron(III), Fe ³⁺

TABLE 5.4.2 Names and formulae of some common anions

Charge		
-1	-2	-3
bromide, Br ⁻ chloride, Cl ⁻ fluoride, F ⁻ iodide, I ⁻	oxide, O ²⁻ sulfide, S ²⁻	nitride, N ³⁻

Rules for writing chemical formulae

Here are some simple rules to follow when you are writing chemical formulae.

- Write the symbol for the positively charged ion first.
- Use subscripts to indicate the number of each ion in the formula. Write the subscripts after the ion they refer to.
- If there is just one ion present in the formula, omit the subscript '1'.
- Do not include the charges on the ions in the balanced formula.

These rules are illustrated in Figure 5.4.2.

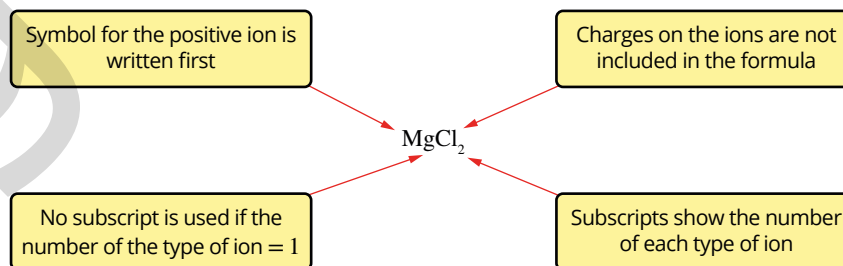


FIGURE 5.4.2 This diagram summarises the information provided by a chemical formula.

Worked example 5.4.1

STEPS IN WRITING A CHEMICAL FORMULA

Determine the chemical formula of the ionic compound formed between zinc and nitride ions. You may need to refer to Tables 5.4.1 and 5.4.2.	
Thinking	Working
Write the symbol and charge of the two ions forming the ionic compound.	Zn^{2+} and N^{3-}
Calculate the lowest common multiple of the two numbers in the charges of the ions.	$2 \times 3 = 6$
Calculate how many positive ions are needed to equal the lowest common multiple.	Three Zn^{2+} ions
Calculate how many negative ions are needed to equal the lowest common multiple.	Two N^{3-} ions
Use the answers from the previous two steps to write the formula for the ionic compound. Write the symbol of the positive ion first. (Note that 1 is not written as a subscript.)	Zn_3N_2

Worked example: Try yourself 5.4.1

STEPS IN WRITING A CHEMICAL FORMULA

Determine the chemical formula of the ionic compound formed between barium and fluoride ions. You may need to refer to Tables 5.4.1 and 5.4.2.

5.4 Review

SUMMARY

- When considering the formula of an ionic compound, the total number of positive charges on the metal ions must equal the total number of negative charges on the non-metal ions.
- When writing formulae of ionic compounds:
 - the symbol for the positively charged ion is written first
 - subscripts are used to indicate the number of each ion in the formula
 - the charges on the ions are not included in the balanced formula as they cancel out.

KEY QUESTIONS

- A_2B is an ionic compound. Both of the ions in A_2B have the same electron configuration as an argon atom. What is the identity of A_2B ?
 - calcium chloride
 - potassium sulfide
 - calcium sulfide
 - sodium oxide
- Listed below are pairs of metal and non-metal ions. When the ions in each pair combine to form ionic compounds, what will be the ratio of metal ion to non-metal ion?
 - K^+ and S^{2-}
 - Al^{3+} and F^-
 - Ca^{2+} and N^{3-}
 - Al^{3+} and P^{3-}
 - Mg^{2+} and Cl^-
- Use the information in Tables 5.4.1 and 5.4.2 on page xx to answer Questions 3 and 4.
 - sodium chloride
 - potassium bromide
 - zinc chloride
 - potassium oxide
 - barium bromide
 - aluminium iodide
 - silver bromide
 - zinc oxide
 - barium oxide
 - aluminium sulfide
- Name the ionic compounds with the following formulae.
 - KCl
 - CaO
 - MgS
 - K_2O
 - NaF

5.5 Writing formulae of more complex ionic compounds

The chemical formulae you have written for ionic compounds so far contain simple ions—ions that contain only one atom of an element. In this section, you will look at the formulae of some compounds containing more complex ions. You will also learn rules to help you name ionic compounds.

FORMULAE CONTAINING POLYATOMIC IONS

Simple ions contain only one atom, as you have seen in section 5.4. However, other ions contain two or more atoms, which may be of different elements. These ions are called **polyatomic ions**.

In polyatomic ions:

- if different elements are present, then they are combined in a fixed ratio
- the group of atoms behaves as a single unit with a specific charge
- subscripts are used to indicate the number of each kind of atom in the ion.

For example, a carbonate ion (CO_3^{2-}) contains one carbon atom and three oxygen atoms combined together to form an ion. The carbonate ion has a charge of -2 . Other polyatomic ions are nitrate (NO_3^-), hydroxide (OH^-) and phosphate (PO_4^{3-}). The formulae of a number of polyatomic ions can be seen in Table 5.5.1.

TABLE 5.5.1 Common polyatomic cations and anions

Charge			
+1	-1	-2	-3
ammonium, NH_4^+	cyanide, CN^- dihydrogenphosphate, H_2PO_4^- ethanoate (acetate), CH_3COO^- hydrogencarbonate, HCO_3^- hydrogensulfate, HSO_4^- hydroxide, OH^- nitrate, NO_3^- nitrite, NO_2^- permanganate, MnO_4^-	carbonate, CO_3^{2-} chromate, CrO_4^{2-} dichromate, $\text{Cr}_2\text{O}_7^{2-}$ hydrogenphosphate, HPO_4^{2-} oxalate, $\text{C}_2\text{O}_4^{2-}$ sulfate, SO_4^{2-} sulfite, SO_3^{2-}	phosphate, PO_4^{3-}

i The common name for the hydrogencarbonate ion is bicarbonate.

If more than one polyatomic ion is required in a formula to balance the charge, then it is placed in brackets with the required number written as a subscript after the brackets. Some examples are:

- magnesium nitrate, $\text{Mg}(\text{NO}_3)_2$
- aluminium hydroxide, $\text{Al}(\text{OH})_3$
- ammonium sulfate, $(\text{NH}_4)_2\text{SO}_4$

Note that brackets are not required for the formula of sodium nitrate (NaNO_3), where there is only one NO_3^- ion present for each sodium ion.

i Polyatomic ions are made up of two or more atoms charge. They need to be written within brackets if there is more than one present in an ionic formula. Subscripts are used to indicate the ratio of ions in the crystal lattice.

A formula that is expressed in terms of the simplest whole-number ratio of particles (in this case the particles are ions) is called an **empirical formula**.

The formulae of several other ionic compounds containing polyatomic ions are shown in Figure 5.5.1.

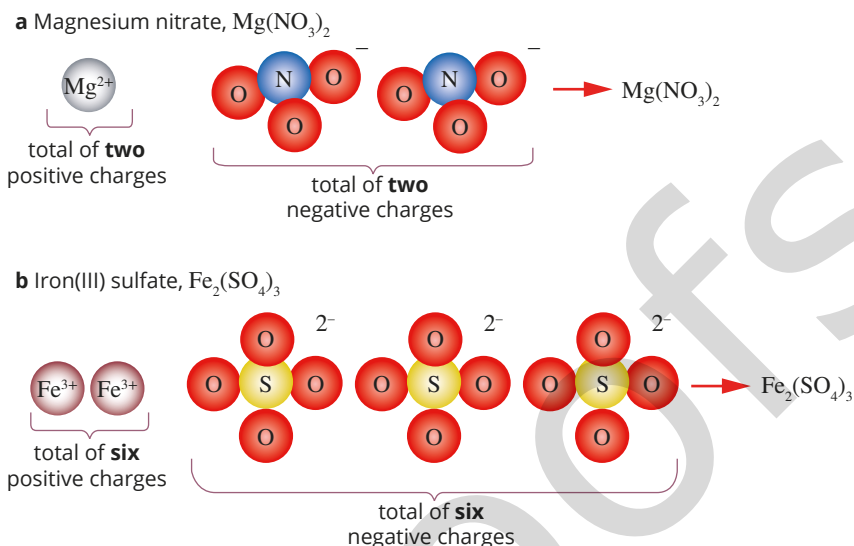


FIGURE 5.5.1 The chemical formulae of ionic compounds containing polyatomic ions

Formulae involving elements with multiple valencies

The **valency** of an atom is a measure of its combining ability with other atoms. For main group metals the valency is equal to the numeral of the ion's charge. Some transition metals however, can form ions with different charges. For example, copper can form Cu^+ ions with a charge of +1 and Cu^{2+} ions with a charge of +2.

Some other metals with variable valency are:

- lead (Pb^{2+} and Pb^{4+})
- iron (Fe^{2+} and Fe^{3+})
- tin (Sn^{2+} and Sn^{4+}).

For compounds of these metals, you need to specify the charge on the ion when naming the compound. This is done by placing a Roman numeral immediately after the metal in the name of the compound. For example:

- iron(II) chloride contains the Fe^{2+} ion and so the formula is FeCl_2
- iron(III) chloride contains the Fe^{3+} ion and so the formula is FeCl_3
- copper(I) sulfide contains the Cu^+ ion and so the formula is Cu_2S .

NAMING IONIC COMPOUNDS

There are some basic conventions that are followed when naming ionic compounds.

- The name of a positively charged ion (cation) remains unchanged. For example, the cation of a sodium atom is called a sodium ion; the cation of an aluminium atom is an aluminium ion.
- For simple non-metal ions (anions), the name of the ion is similar to that of the atom, but ends in '-ide'. For example, the anion of the chlorine atom is chloride; the anion of the oxygen atom is oxide.
- For polyatomic anions containing oxygen, the name of the ion will usually end in '-ite' or '-ate'. For example, the NO_2^- ion is called a nitrite ion; the NO_3^- ion is called a nitrate ion. (For two different ions of the same element with oxygen, the name of the ion with the smaller number of oxygen atoms usually ends in '-ite' and the one with the larger number of oxygen atoms ends in '-ate'.)

i Some transition metals can form ions with variable charges. To indicate the charge on the metal, write a Roman numeral in brackets immediately after the metal in the name of the compound.

5.5 Review

SUMMARY

- When writing formulae for ionic compounds, the following rules apply.
 - Ions that contain two or more atoms of different elements are called polyatomic ions.
 - If a chemical formula contains more than one polyatomic ion, the formula of the ion is placed in brackets with the number of ions written as a subscript after the brackets.
- When naming ionic compounds, the following rules apply.
 - The name of the metal ion is the same as the name of the metal.
 - Simple non-metal ions take the name of the atom, but end in '-ide'.
 - Polyatomic anions containing oxygen usually end in '-ite' or '-ate'.
 - For metals that form ions with different charges, the charge on the ion is shown by placing a Roman numeral after the name of the metal.
- The valency of a metal is equal to the charge of its ions.

KEY QUESTIONS

Use the information in Tables 5.4.1 and 5.4.2 on page xx and Table 5.5.1 on page xx to answer these questions.

1 Write formulae for the following ionic compounds.

- a** sodium carbonate
- b** barium nitrate
- c** aluminium nitrate
- d** calcium hydroxide
- e** zinc sulfate
- f** potassium hydroxide
- g** potassium nitrate
- h** zinc carbonate
- i** potassium sulfate
- j** barium hydroxide

2 Write formulae for the following ionic compounds.

- a** caesium chloride
- b** iron(III) oxide
- c** copper(II) oxide
- d** chromium(III) sulfate
- e** iron(II) oxide
- f** aluminium oxide
- g** calcium nitrate

3 Name the following ionic compounds.

- a** $\text{Mg}(\text{OH})_2$
- b** Na_2CO_3
- c** FeSO_4
- d** CuSO_4
- e** $\text{Ba}(\text{NO}_3)_2$
- f** Cu_2SO_4
- g** $\text{Fe}_2(\text{SO}_4)_3$
- h** NH_4NO_3
- i** Na_2HPO_4

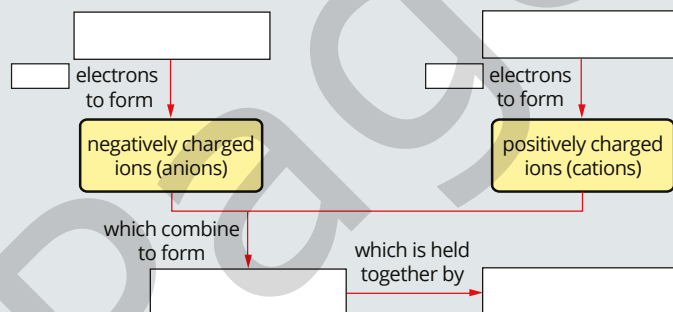
Chapter review

KEY TERMS

chemical formula	ionic bonding
crystal lattice	ionic compound
electrolyte	polyatomic ion
electron transfer diagram	valency
empirical formula	

Properties and structures of ionic compounds

- Describe an experiment you could carry out to demonstrate each of the following properties of the compounds given. In each case:
 - sketch the equipment you would use
 - describe what you would expect to observe.
 - Solid magnesium chloride does not conduct electricity.
 - Molten sodium chloride does conduct electricity.
 - Solid sodium chloride is hard and brittle.
- State whether the following properties relate to metallic lattices only, ionic lattices only or both metallic and ionic lattices.
 - They contain both positively and negatively charged particles.
 - The lattice is held together by forces of attraction between positively and negatively charged particles.
 - They are hard.
 - They are brittle.
 - They conduct electricity in the molten state.
- Copy and complete the following diagram to show what happens when ionic compounds are formed from metallic and non-metallic elements.



Using the ionic bonding model to explain properties

- Use the ionic bonding model to explain the following properties of ionic compounds.
 - They generally have high melting points.
 - They are hard and brittle.
 - They do not conduct electricity in the solid state but will conduct when molten or dissolved in water.

- The melting point of sodium chloride is 801°C , whereas that of magnesium oxide is 2800°C .
 - What particles are present in the two solids?
 - Which solid has the stronger forces between its particles?
 - Give some possible differences in the structure and bonding of the two solids that would explain the large difference in melting points.
- Describe what happens to the forces between particles as sodium chloride is heated and melts.
- The following is a list of some of the uses of ionic compounds. For each use, explain what property of ionic compounds enables them to be used in this way.
 - insulators on electrified fences
 - bricks for building the wall of a house
 - one of the chemicals in a battery

Formation of ionic compounds

- Use diagrams to show the electron transfer that occurs when:
 - lithium reacts with chlorine
 - magnesium reacts with fluorine
 - potassium reacts with sulfur
 - magnesium reacts with nitrogen.
- Give the electron configurations of the following ions.
 - Na^{+}
 - O^{2-}
 - Mg^{2+}
 - N^{3-}
- The electron configurations of some metallic and non-metallic elements are given. (The symbols shown for the elements are not their real ones.) Write formulae for the compounds they are most likely to form if they react together. The first example has been done for you.

a A: 2,1	B: 2,6	A_2B
b C: 2,8,3	D: 2,7	
c E: 2,8,8,2	F: 2,8,6	
d G: 2,8,8,1	H: 2,5	
e K: 2,8,2	L: 2,6	

- 11** Refer to the periodic table at the end of the book and, for each general formula given, identify two elements that will react to form an ionic compound with that formula. (Remember the metal ion, as represented by X, is written first in each formula.)
- XY_2
 - XY
 - X_2Y
 - X_3Y
 - XY_3
 - X_3Y_2
- 12** Explain why elements in group 17 of the periodic table are likely to form ions with a -1 charge.
- 13** Write an equation for the reaction between the following metal and non-metal atoms. Show the electron configurations for each element before and after the reaction.
- magnesium and chlorine atoms
 - aluminium and oxygen atoms

Chemical formulae of simple ionic compounds

- 14** Write the empirical formula for the ionic compound formed in the reaction between:
- potassium and bromine
 - magnesium and iodine
 - calcium and oxygen
 - aluminium and fluorine
 - calcium and nitrogen.
- 15** Write the empirical formula for each of the following ionic compounds.
- copper(I) chloride
 - silver oxide
 - lithium nitride
 - potassium iodide
- 16** What do subscripts in the formula of an ionic compound tell you about the arrangement of the metal and non-metal ions?

Writing formulae of more complex ionic compounds

- 17** The elements X, Y and Z form ionic compounds when they react with other elements. The following compounds are formed: Ca_3X_2 , Y_2CO_3 and Al_2Z_3 .
- What is the charge of the ion formed by:
 - element X?
 - element Y?
 - element Z?

- Use these charges on the ions to write correct chemical formulae for the:
 - sulfate salt of Y
 - potassium salt of Z
 - ionic compound formed between X and Y
 - ionic compound formed between Y and Z.

- 18** Write the empirical formulae for the following ionic compounds.
- copper(I) nitrate
 - chromium(II) fluoride
 - potassium carbonate
 - magnesium hydrogencarbonate
 - nickel(II) phosphate
- 19** Name the ionic compounds with the following chemical formulae.
- $(\text{NH}_4)_2\text{CO}_3$
 - $\text{Cu}(\text{NO}_3)_2$
 - CrBr_3

Connecting the main ideas

- 20** A student compares the structure and bonding in metals and ionic compounds and makes the following statements.
- Metals and ionic solids both contain positive ions in a regular arrangement.
 - In metals and ionic solids, there is attraction between one particle and all the neighbouring particles of opposite charge.
 - In metals and ionic solids, there will be forces of repulsion between particles with like charges.
 - In metals and ionic solids that contain metal cations, energy has been used to remove valence electrons from metal atoms.
 - A metal will conduct electricity, whereas an ionic solid will not because electrons are much smaller than negative ions.
- Comment on each of these statements, explaining clearly why you either agree or disagree.
- 21** Construct a concept map to show the connection between these terms: metal, non-metal, atom, valence electron, anion, cation, electrostatic attraction, ionic bonding.

